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Department of Artificial Intelligence
Intelligent Multi-Agent and Expert Systems (AI401), Semester VII

Instructions

1. Save your file as LAB_No_Roll_No.doc (e.g. LAB_3_U21AI001.doc).
2. Paste a snapshot of each step followed along with the result(s).
3. Record your observations and comments after the results.

Experiment 3

Agent Architectures — The BDI Deliberation Cycle

Aim

To implement a Belief–Desire–Intention agent that revises its beliefs about a dynamic environment, deliberates over competing desires to commit to an intention, and executes plans under a chosen commitment strategy, and to measure how that commitment affects performance as the world changes faster.

Theory

A reactive agent maps percepts straight to actions; a deliberative agent reasons about its goals first. The BDI architecture formalises that reasoning with three attitudes. Beliefs are what the agent takes to be true, revised from percepts. Desires are the states of affairs it would like to bring about — options, not commitments, and they may conflict. Intentions are the desires it has committed to and is pursuing. An intention is more than a preference: it constrains future deliberation and persists until it is achieved, becomes impossible, or is dropped.

The architecture is a control loop over four functions: brf(B, p) folds a percept into the belief set, options(B, I) proposes desires consistent with current beliefs and intentions, filter(B, D, I) is deliberation proper and returns the intention committed to, and plan(B, I) performs means–ends reasoning to produce the actions that achieve it. Two of these are familiar: testing whether a plan schema is applicable is the match phase of Experiment 2, and choosing between applicable options is the conflict resolution of Experiment 5. What is new is time pressure — the world does not pause while the agent thinks, so deliberation is itself an action with a cost.

That cost makes commitment a design decision. An agent re-deliberating after every action is perfectly informed but barely acts; one that never reconsiders may spend a whole plan chasing an intention that became unachievable at its second step. Three strategies span the range: blind commitment holds an intention until it is believed achieved, single-minded commitment also drops it once believed unachievable, and open-minded commitment keeps it only while it remains among the best options. The reconsideration interval γ — actions executed between successive deliberations — parameterises the spectrum, $\gamma = 1$ giving a cautious agent and a γ larger than any plan a bold one. Kinny and Georgeff showed that bold agents dominate in slowly changing worlds and that the advantage passes to cautious agents as the rate of change rises.

The testbed is a simplified Tileworld: a 15 × 15 grid on which holes appear at random cells with fixed probability per world tick and expire after a random lifetime, the agent scoring by moving onto a hole and filling it. Effectiveness is the ratio of holes filled to holes that appeared, and world dynamism is set by the number of world ticks per agent action. Holes are the desires, the hole being pursued is the intention, and the shortest path to it is the plan.

Algorithm

Algorithm 1 — The BDI deliberation cycle

1. Take the first percept p, set B := brf(∅, p), and initialise I := null and π := empty.
2. Obtain the current percept p and revise beliefs, B := brf(B, p).

3. If I is null or π is empty, generate the options D := options(B, I), commit by I := filter(B, D, I) and construct π := plan(B, I) by means–ends reasoning. If D is empty, take no action this step and let the world advance.
4. Execute the first action of π and delete it from π. The world advances by its own clock while the action is performed.
5. If I is achieved in the revised beliefs, drop I and π and return to step 2.
6. Otherwise invoke Algorithm 2 to decide whether to reconsider I, then return to step 2, repeating for the required number of agent time steps.

Algorithm 2 — Intention reconsideration under single-minded commitment

1. Maintain a counter s of the actions executed since the last reconsideration and increment it after every action taken in step 4 of Algorithm 1.
2. If $s < \gamma$, return immediately with I and π unchanged: the agent stays committed and does not deliberate.
3. Otherwise set s := 0 and charge one agent time step to deliberation, allowing the world to advance while the agent reasons.
4. Re-perceive and revise beliefs. If I is no longer achievable under the revised beliefs, drop I and π and return with no intention, so that the next cycle commits afresh.
5. Re-run options and filter on the revised beliefs. If the preferred option differs from I, adopt it as the new intention and re-plan; otherwise retain I and π and continue executing the existing plan.
6. Report γ with the results: $\gamma = 1$ yields a cautious agent and a γ exceeding the longest plan yields a bold one.

Procedure and Result

Paste the snapshots of the steps followed and the output produced by the BDI agent below, including the event trace showing where the agent commits to an intention, achieves one, drops one that has become unachievable, and switches to a better option on reconsideration, together with the holes filled, the holes that appeared, and the effectiveness obtained for the value of γ used.

Observations

Comment on what the drop and switch events reveal about the persistence of intentions and about the difference between single-minded and open-minded commitment; on why deliberation must be charged a time step for the comparison to be meaningful; and on which value of γ gave the highest effectiveness and how it moved as the world was made to change faster.

Exercises

1. Boldness against dynamism. Sweep γ over {1, 2, 4, 8, bold} and the world speed over {1, 2, 4, 8} world ticks per agent action, averaging effectiveness over at least 25 random seeds of 600 agent steps each. Tabulate and plot the results, state the value of γ that maximises effectiveness at each speed, and explain why that optimum falls as the world speeds up and why both extremes — reconsidering after every action and never reconsidering at all — are beaten by an intermediate value.
2. A resource-bounded filter. Give the agent a battery of capacity 40 that loses one unit per move and a charger cell that restores it, and build a plan library of two schemas: fill-hole, applicable only when the battery covers the trip plus the return to the charger, and recharge, always applicable. Have the filter commit to the nearest hole whose precondition holds and recharge otherwise. Compare against a battery-blind agent that always chases the nearest hole, reporting holes filled and steps spent stranded, and state which BDI component the fix belongs in and why not the planner.